Azure SQL Database maintenance essentials

Sergio Govoni

Slide and demo: bit.ly/global-azure-2026-sgovoni

UNIPORDENONE
moltiplica il valore della formazione

TD SYNNEX

About me

linkedin.com/in/sgovoni

github.com/segovoni

twitter.com/segovoni

Sergio Govoni

Microsoft® Most Valuable Professional

Agenda

- Azure SQL Database maintenance tasks
  - Required activities
  - Methods
  - Azure SQL Elastic Jobs

The starting point

- Azure SQL Database hasn’t got a native scheduling service like SQL Agent of the on-premise instances
- When a database(s) is deployed in Azure SQL Database, possible solutions to perform maintenance activities must be studied!

Microsoft responsibility

- Microsoft is responsible for the following’s tasks:
  - Backup
  - High availability
  - SQL Server updates
  - Windows updates
  - Consistency checks
  - Performance monitoring

Automatic updates

- No SQL update required
- No OS update required
- Microsoft has committed to keeping the SQL Engine unchanged within a specific compatibility level
- Make sure your application is aligned with the compatibility level configured for your database

Azure SQL planned maintenance

- Azure performs planned maintenance to keep Azure SQL secure, compliant, stable, and performant
- Most updates are transparent thanks to technologies such as hotpatching, but some maintenance events can cause short service interruptions
- During maintenance, a reconfiguration/failover can occur: the primary replica moves to another replica or compute node, and applications must reconnect

Azure SQL planned maintenance

- Reconfigurations/failovers generally complete within 30 seconds, the average is 8 seconds
- Long-running queries can be interrupted, and new connections may receive error 40613 - Database Unavailable
- Applications should include retry logic to handle transient connectivity errors
- On average, 1 planned maintenance event occurs each 35 days

What remaining tasks do I have in Azure SQL?

- Performance tuning (to save your company money)
- User management and security
- Configure disaster recovery across Azure regions
- Typical DBA tasks include index maintenance, statistics updates, migrations, monitoring, and other routine database operations

Backups

Backups

- Azure SQL Database automatically creates database backups
  - Full backups every week
  - Differential backups every 12 hours
  - Transaction log backups every 5-10 minutes
- Backups are stored in read-access geo-redundant storage replicated to a paired data center

Backups can be used for..

- Restore
  - Existing database to a point-in-time
  - Deleted database to the time it was deleted
  - Database to another geographical region
  - Database from a specific long-term backup
  - https://bit.ly/34Y1IY2

Backups retention

- All Azure SQL databases have a default backup retention period of **7** days
- Backup retention period can be changed up to **35** days (at 7-days chunks)
- Backups of deleted databases will be kept according to the retention period fixed
- https://bit.ly/2SpEOq0

Backups long-term retention (LTR)

- If you want to keep your backups for a period of time longer than the maximum allowed, you can evaluate the long-term backup retention service
- By using long-term retention, you can store specific backups up to **10** years
- https://bit.ly/2EOPVkr

Maintenance activities

Maintenance activities

- The key maintenance activities for database solutions implemented on Azure SQL are
  - Query tuning
  - Index maintenance
  - Statistics update

General methods for DB maintenance activities

- The typical methods to do maintenance activities are:
  - Database Maintenance Plans
  - Ola Hallengren's Maintenance Solution Suite
  - Use of DBATools with PowerShell
- On-premise maintenance activities are scheduled with:
  - SQL Agent
  - Windows Scheduled Task
  - Cron in Linux environment
  - Third parties' tools

Methods for Azure SQL DB maintenance activities

- In Azure SQL Database we have several methods to do maintenance operations, this is the complete list:
  - Linked Server
  - Database Maintenance Plan
  - Azure Automation Services
  - Azure SQL Database Elastic Jobs
- It is advised to use the stored procedures of the suite SQL Server Maintenance Solution done by Ola Hallengren
  - https://ola.hallengren.com

Automatic index compaction (preview)

- It helps you to reduce the consumption of
  - Storage space
  - Disk I/O
  - CPU
  - Memory
- It improves workload performance without investing time and effort into index maintenance jobs
- It is disabled by default

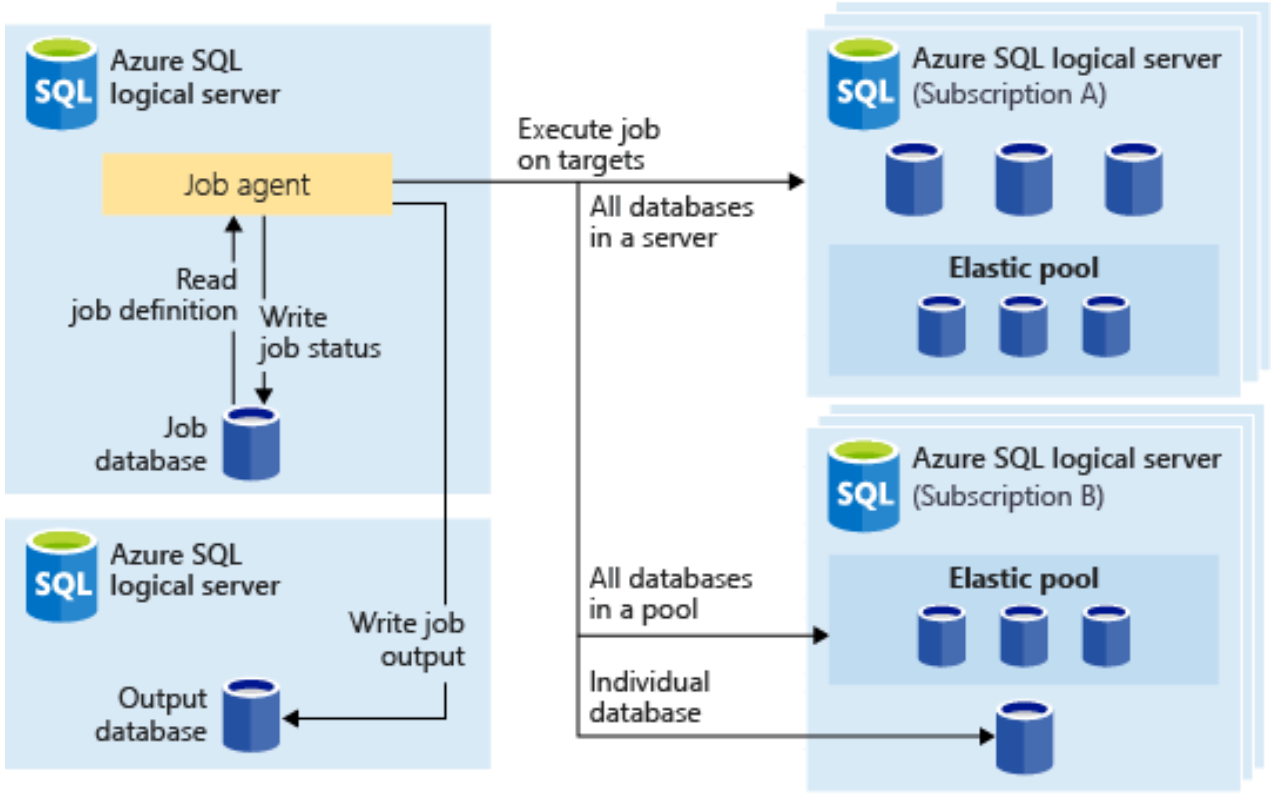
Automatic index compaction (preview)

- Benefits and considerations
  - You don't need to set up and maintain index maintenance jobs
  - It avoids high resource consumption by index maintenance jobs
  - It reduces the space growth that can occur when data in the database is modified
  - It improves query performance.
    - Reading a compact index reads fewer pages
    - A compact index is more likely to be selected to improve a query plan
  - Similar to index reorganization, the compaction process acquires short-term exclusive (X) page locks

Azure SQL Database Elastic Jobs

The most important part of the session is going to start

Azure SQL Database Elastic Jobs



https://learn.microsoft.com/azure/azure-sql/database/elastic-jobs-overview?view=azuresql

Azure SQL Database Elastic Jobs

- Azure SQL Database Elastic Jobs is the **most complete** solution for **automation** of scheduled activities to be performed on Azure SQL Database
- It’s in general availability
- It runs jobs whose definitions are stored in a “job database”
- Jobs are T-SQL scripts scheduled to act for a group of Azure SQL databases

When to use Elastic Jobs

- Automate **management tasks** and **schedule** them
- Configure jobs to execute across a **collection** of databases on a recurring basis, such as during off-peak hours
  - This is a significant differentiation from SQL Agent, which cannot dynamically enumerate the target databases, especially in SaaS customer scenarios where databases are added/deleted dynamically
- Collect data for reporting
- Data movement

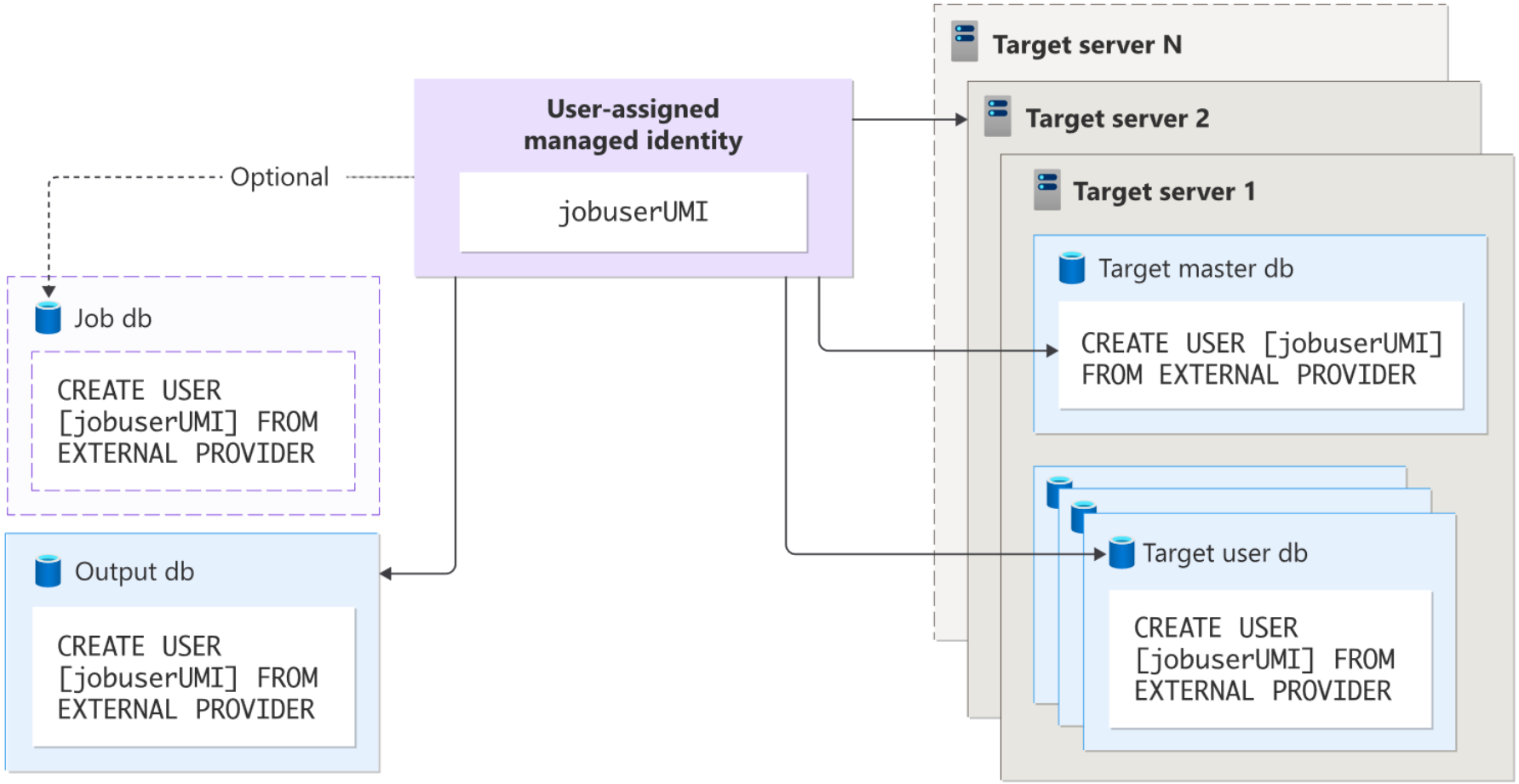
The job database

- Azure SQL Database Elastic Jobs requires an empty database with service level of "**S1**" or higher hosted in an Azure SQL Database
  - This DB is also known as "job database" and it's like the MSDB for on-premises instance
  - The "job database" is used for defining jobs and tracking the status and history of job executions

Elastic jobs authentication

- The elastic job agent can connect to the server(s)/database(s) specified by the target group via two authentication options:
  - Microsoft Entra (formerly Azure Active Directory) authentication with a user-assigned managed identity (UMI) – Recommended option
  - Database-scoped credentials (previously, this was the only method of authentication)

Microsoft Entra authentication with UMI



DEMO

Limitations and availability

- Azure SQL Team is actively working to remove as many limitations as possible
- Anyway, you can refer to the documentation for a list of known limitations, here: https://aka.ms/EJ-limitations
- Elastic Jobs is available in all the regions that Azure SQL Database is available

Summary and Resources

Summary

- Database maintenance is an important factor also for Azure SQL
- Azure SQL Database is a PaaS solution, Microsoft is responsible for the infrastructure, backup, HA, updates, consistency checks, performance monitoring
  - Database maintenance is under the DBA responsibility
- Elastic Jobs is a fully integrated Azure SQL Database service that allows you to automate and manage administrative tasks across multiple SQL databases in a secure and scalable way
- Jobs can be run on a schedule or on-demand, targeting any tier of Azure SQL Database

Resources

- Automate management tasks in Azure SQL
- Elastic jobs in Azure SQL Database
- Create, configure, and manage elastic jobs
- Troubleshooting common issues with Elastic Jobs in Azure SQL Database
- Automatic index compaction (preview)
- Automating Azure SQL Database maintenance tasks
  - 1° Part: https://bit.ly/3clqzJE
  - 2° Part: https://bit.ly/2wOpJG6
  - 3° Part: https://bit.ly/3VYBDGC

BY COMMUNITY • FOR COMMUNITY •
#GLOBALAZURE
2026

Q&A

inn0va
community

TD SYNNEX

UNIPORDENONE
moltiplica il valore della formazione

BY COMMUNITY • FOR COMMUNITY
#GLOBALAZURE
2026

Thanks!

inn0va
community

TD SYNNEX

UNIPORDENONE
moltiplica il valore della formazione